Time-reversal refocusing in homogeneous and randomly heterogeneous media

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1 Introduction

Objective The objective of this project is to demonstrate the properties of time reversal within heterogeneous media. Specifically, it highlights how wave propagation through a random medium can lead to a super-resolution effect during the refocusing process, effectively overcoming the classical diffraction limit. This phenomenon provides significant advantages for high-precision applications, such as the non-invasive destruction of kidney stones.

Structure of the report This document provides the theoretical framework and detailed mathematical derivations of the project. It serves as a core component of a comprehensive submission, which also includes:

- A Jupyter Notebook showcasing numerical results and simulations.
- A Python library, developed specifically for this project, which contains the core functions called throughout the notebook.
- A .mp4 animation illustrating the dynamic refocusing process described in the final section.

2 Preliminaries

This section establishes the mathematical framework and the notation used throughout the project. Let u denote the time-dependent wave function. To simplify the analysis, we perform a Fourier transform to decompose the signal into its spectral components. This allows us to focus on monochromatic waves in the frequency domain. The time-domain solution is subsequently reconstructed by synthesizing these individual components through an inverse Fourier transform.

Our domain is described by the function c. Let $u : (\mathbf{x}, t) \in \mathbb{R}^2 \times [0, +\infty) \to \mathbb{R}$ be a scalar wave field. It satisfies the wave equation

$$\left(\frac{1}{c(\mathbf{x})^2}\partial_t^2 - \Delta\right) u(\mathbf{x}, t) = 0.$$

In order to solve this equation, we can use the Fourier transform of u defined by

$$\hat{u} : (\mathbf{x}, \omega) : \mathbb{R}^2 \times [0, +\infty) \mapsto \int_{\mathbb{R}} u(\mathbf{x}, t) e^{it\omega} dt$$

and then reconstruct the function u using the inverse Fourier transform

$$\forall \mathbf{x} \in \mathbb{R}^2,\ \forall t \in [0, +\infty),\ u(\mathbf{x}, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{u}(\mathbf{x}, \omega) e^{-it\omega} d\omega.$$

In the frequency domain, we obtain the corresponding Helmholtz equation

$$\left(\frac{\omega^2}{c(\mathbf{x})^2} + \Delta\right) \hat{u}(\mathbf{x}, \omega) = 0. \tag{1}$$

3 Paraxial approximation in a homogeneous medium

In this section, we assume a homogeneous medium defined by a constant velocity $c(\mathbf{x}) = c_0$. Let us consider a fixed frequency ω and denote the spatial field by $\hat{u}_\omega(\mathbf{x}) = \hat{u}(\mathbf{x}, \omega)$. We introduce the function $\phi(\mathbf{x})$ such that the field can be factored as a modulated plane wave propagating along the z-axis: $\hat{u}_\omega(\mathbf{x}) = \phi(\mathbf{x})e^{ikz}$, where $k = \omega/c_0$.

We can now derive the differential equation governing ϕ. Substituting into (1) yields

$$\left(k^2\phi(\mathbf{x}) + \partial_x^2\phi(\mathbf{x}) + \partial_z^2\phi(\mathbf{x}) + 2ik\partial_z\phi(\mathbf{x}) - k^2\phi(\mathbf{x})\right) e^{ikz} = 0,$$

which simplifies to

$$\Delta\phi(\mathbf{x}) + 2ik\partial_z\phi(\mathbf{x}) = 0. \tag{2}$$

To proceed, we apply the paraxial approximation. We assume that $|\partial_z^2\phi| \ll |2k\partial_z\phi|$. Under this assumption, equation (2) reduces to the paraxial Schrödinger equation

$$2ik\partial_z\phi(\mathbf{x}) + \partial_x^2\phi(\mathbf{x}) = 0. \tag{3}$$

We consider a time-harmonic initial condition in the plane $z = 0$ with a transverse Gaussian profile of radius r_0

$$\forall x \in \mathbb{R}, \quad \phi(z = 0, x) = \phi_0(x) = \exp\left(-\frac{x^2}{r_0^2}\right).$$

We aim to solve the following problem:

$$\begin{cases} \partial_z\phi(\mathbf{x}) = \frac{i}{2k}\partial_x^2\phi(\mathbf{x}) & \forall \mathbf{x} \in \mathbb{R}^2 \\ \phi(z = 0, x) = \phi_0(x) & \forall x \in \mathbb{R}. \end{cases}$$

Applying the Fourier transform with respect to the transverse variable x transforms the partial differential equation into a set of ordinary differential equations in the spectral domain

$$\begin{cases} \partial_z\hat{\phi}(z, \kappa) = -\frac{i\kappa^2}{2k}\hat{\phi}(z, \kappa) & \forall (z, \kappa) \in \mathbb{R}^2 \\ \hat{\phi}(z = 0, \kappa) = \hat{\phi}_0(\kappa) = r_0\sqrt{\pi}e^{-\frac{\kappa^2 r_0^2}{4}} & \forall \kappa \in \mathbb{R}. \end{cases}$$

For a fixed $\kappa \in \mathbb{R}$, the function $z \mapsto \hat{\phi}(z, \kappa)$ satisfies a simple first-order linear ODE. By the Cauchy-Lipschitz theorem, it admits a unique solution

$$\hat{\phi}(z, \kappa) = r_0\sqrt{\pi}\exp\left(-\kappa^2\left(\frac{r_0^2}{4} + \frac{iz}{2k}\right)\right).$$

Finally, we recover the spatial profile using the inverse Fourier transform. The resulting field is

$$\forall (z, x) \in \mathbb{R}^2, \quad \phi(z, x) = \frac{r_0}{\sqrt{r_0^2 + \frac{2iz}{k}}}\exp\left(-\frac{x^2}{r_0^2 + \frac{2iz}{k}}\right).$$

To match standard notation, we define $r_t = r_0\sqrt{1 + \frac{2iL}{kr_0^2}}$. The transmitted wave profile at $z = L$, denoted by $\phi_t(x) = \phi(L, x)$, is then given by

$$\phi_t(x) = \frac{r_0}{r_t}\exp\left(-\frac{x^2}{r_t^2}\right).$$

4 Time reversal for time-harmonic waves in a homogeneous medium

We consider a time-reversal mirror located in the plane $z = L$. We first examine a compact mirror given by

$$\chi_M(x) = \left(1 + \left(\frac{x}{2r_M}\right)^2\right)^2 \mathbf{1}_{[-2r_M, 2r_M]}(x).$$

The refocused wave ϕ^{tr} is the solution to the following boundary value problem:

$$\begin{cases} \partial_z \phi^{\mathrm{tr}}(\mathbf{x}) = \frac{i}{2k}\partial_x^2 \phi^{\mathrm{tr}}(\mathbf{x}) & \forall \mathbf{x} \in \mathbb{R}^2, \\ \phi^{\mathrm{tr}}(L, x) = \overline{\phi_t(x)}\chi_M(x) & \forall x \in \mathbb{R}. \end{cases}$$

While an analytical solution is not available for the general case, we can derive an exact solution if we take a Gaussian mirror

$$\chi_M(x) = \exp\left(-\frac{x^2}{r_M^2}\right).$$

First, we compute the initial condition for the time-reversal problem at $z = L$. Using the expression of the transmitted wave ϕ_t, we have

$$\begin{aligned} \phi^{\mathrm{tr}}(L, x) &= \overline{\phi_t(x)}\chi_M(x) \\ &= \frac{r_0}{\overline{r_t}} \exp\left(-\frac{x^2}{\overline{r_t}^2}\right) \exp\left(-\frac{x^2}{r_M^2}\right) \\ &= \frac{r_0}{\overline{r_t}} \exp\left(-x^2\left(\frac{r_M^2 + \overline{r_t}^2}{r_M^2 \overline{r_t}^2}\right)\right). \end{aligned}$$

Applying the Fourier transform with respect to x yields

$$\begin{aligned} \widehat{\phi^{\mathrm{tr}}}(L, \kappa) &= \frac{r_0}{\overline{r_t}} \sqrt{\frac{\pi r_M^2 \overline{r_t}^2}{r_M^2 + \overline{r_t}^2}} \exp\left(-\frac{\kappa^2}{4} \frac{r_M^2 \overline{r_t}^2}{r_M^2 + \overline{r_t}^2}\right) \\ &= \frac{r_0 r_M \sqrt{\pi}}{\sqrt{r_M^2 + \overline{r_t}^2}} \exp\left(-\frac{\kappa^2 r_M^2 \overline{r_t}^2}{4(r_M^2 + \overline{r_t}^2)}\right). \end{aligned}$$

We now solve the propagation equation in the Fourier domain

$$\partial_z \widehat{\phi^{\mathrm{tr}}}(z, \kappa) = -\frac{i\kappa^2}{2k} \widehat{\phi^{\mathrm{tr}}}(z, \kappa).$$

Integrating from the mirror plane $z = L$ to the observation distance z, we obtain

$$\widehat{\phi^{\mathrm{tr}}}(z, \kappa) = \widehat{\phi^{\mathrm{tr}}}(L, \kappa) \exp\left(-\frac{i\kappa^2(z - L)}{2k}\right).$$

Substituting the initial condition, it comes

$$\widehat{\phi^{\mathrm{tr}}}(z, \kappa) = \frac{r_0 r_M \sqrt{\pi}}{\sqrt{r_M^2 + \overline{r_t}^2}} \exp\left(-\frac{\kappa^2}{4}\left[\frac{r_M^2 \overline{r_t}^2}{r_M^2 + \overline{r_t}^2} + \frac{2i(z - L)}{k}\right]\right).$$

We are interested in the refocused field at $z = 2L$. For conciseness, we define the complex width r_{tr} at this distance such that

$$r_{\mathrm{tr}}^2 = \frac{r_M^2 \overline{r_t}^2}{r_M^2 + \overline{r_t}^2} + \frac{2iL}{k}.$$

The solution at $z = 2L$ simplifies to

$$\widehat{\phi^{\mathrm{tr}}}(2L, \kappa) = \frac{r_0 r_M \sqrt{\pi}}{\sqrt{r_M^2 + \overline{r_t}^2}} \exp\left(-\frac{\kappa^2 r_{\mathrm{tr}}^2}{4}\right).$$

Finally, applying the inverse Fourier transform provides the spatial profile of the refocused wave

$$\begin{aligned} \phi^{\mathrm{tr}}(2L, x) &= \frac{r_0 r_M \sqrt{\pi}}{\sqrt{r_M^2 + \overline{r_t}^2}} \cdot \frac{1}{\sqrt{\pi} r_{\mathrm{tr}}} \exp\left(-\frac{x^2}{r_{\mathrm{tr}}^2}\right) \\ &= \frac{r_0 r_M}{\sqrt{(r_M^2 + \overline{r_t}^2) r_{\mathrm{tr}}^2}} \exp\left(-\frac{x^2}{r_{\mathrm{tr}}^2}\right). \end{aligned}$$

To simplify the amplitude term, we expand the squared denominator $D^2 = (r_M^2 + \overline{r_t}^2) r_{\mathrm{tr}}^2$:

$$\begin{aligned} D^2 &= (r_M^2 + \overline{r_t}^2)\left(\frac{r_M^2 \overline{r_t}^2}{r_M^2 + \overline{r_t}^2} + \frac{2iL}{k}\right) \\ &= r_M^2 \overline{r_t}^2 + \frac{2iL}{k}(r_M^2 + \overline{r_t}^2). \end{aligned}$$

Recalling that $\overline{r_t}^2 = r_0^2 - \frac{2iL}{k}$, we substitute and simplify

$$\begin{aligned} D^2 &= r_M^2\left(r_0^2 - \frac{2iL}{k}\right) + \frac{2iL}{k} r_M^2 + \frac{2iL}{k}\left(r_0^2 - \frac{2iL}{k}\right) \\ &= r_M^2 r_0^2 + \frac{2iL}{k} r_0^2 + \frac{4L^2}{k^2} \\ &= r_M^2 r_0^2 \left(1 + \frac{2iL}{k r_M^2} + \frac{4L^2}{k^2 r_0^2 r_M^2}\right). \end{aligned}$$

Define the attenuation factor a_{tr} as

$$a_{\mathrm{tr}} = \sqrt{1 + \frac{4L^2}{k^2 r_0^2 r_M^2} + \frac{2iL}{k r_M^2}}.$$

The refocused wave profile at $z = 2L$ is finally given by

$$\phi_r^{\mathrm{tr}}(x) = \phi^{\mathrm{tr}}(2L, x) = \frac{1}{a_{\mathrm{tr}}} \exp\left(-\frac{x^2}{r_{\mathrm{tr}}^2}\right).$$

5 Paraxial approximation in a random medium

In a randomly heterogeneous medium, the wave velocity $c(\mathbf{x})$ fluctuates. Introducing the refractive index $n(\mathbf{x}) = c_0/c(\mathbf{x})$, the Helmholtz equation (1) becomes:

$$\left(k^2 n(\mathbf{x})^2 + \Delta\right) \hat{u}(\mathbf{x}, \omega) = 0,$$

where $k = \omega/c_0$ is the background wavenumber. We express the field as a modulated plane wave propagating along the z-axis, $\hat{u}(\mathbf{x}) = \phi(\mathbf{x}) e^{ikz}$. Substituting this into the wave equation yields:

$$\left(k^2 n(\mathbf{x})^2 \phi(\mathbf{x}) + \left(\partial_x^2 + \partial_z^2 + 2ik\partial_z - k^2\right)\phi(\mathbf{x})\right) e^{ikz} = 0.$$

Grouping the terms, this simplifies to:

$$\partial_x^2 \phi(\mathbf{x}) + \partial_z^2 \phi(\mathbf{x}) + 2ik\partial_z \phi(\mathbf{x}) + k^2(n(\mathbf{x})^2 - 1)\phi(\mathbf{x}) = 0.$$

In the paraxial approximation ($|\partial_z^2 \phi| \ll |2k\partial_z \phi|$), we neglect the second-order derivative in the direction of propagation z. The scalar wave equation in a heterogeneous medium with refractive index contrast involves a potential term. As suggested by the subject, we model this heterogeneity by a random function $(z, x) \mapsto \mu(z, x)$, leading to the wave equation

$$2ik\partial_z \phi(z, x) + \partial_x^2 \phi(z, x) + k^2(n(z, x)^2 - 1)\phi(z, x) = 0.$$

Rearranging the terms, we identify the evolution equation for the amplitude ϕ

$$\partial_z \phi = \frac{i}{2k}\partial_x^2 \phi + \frac{ik}{2}\mu(z, x)\phi.$$

5.1 Modeling the random medium

The potential μ represents the random inhomogeneities of the medium. We model the medium as being piecewise constant along the propagation axis z. Specifically, for a longitudinal correlation length z_c, we define

$$\mu(z, x) = \mu_n(x) \quad \text{for } z \in [nz_c, (n+1)z_c),$$

where the sequence $(\mu_n(x))_n$ consists of independent realizations of a transverse Gaussian process with covariance $\mathbb{E}[\mu_n(x)\mu_n(x')] = \sigma^2 \exp(-(x - x')^2/x_c^2)$.

5.2 Numerical resolution: The Split-Step Fourier Method

To solve this equation numerically over a propagation distance L, we discretize the longitudinal axis. The problem can be written as an abstract evolution equation

$$\partial_z \phi = (\hat{D} + \hat{N}(z))\phi,$$

where we define two non-commuting operators

- $\hat{D} = \frac{i}{2k}\partial_x^2$: the linear diffraction operator (diagonal in Fourier space).
- $\hat{N}(z) = \frac{ik}{2}\mu(z, \cdot)$: the refraction operator (diagonal in real space).

We solve this equation iteratively. Let us focus on a single interval $[nz_c, (n+1)z_c]$. Under the piecewise constant assumption on μ, the operator $\hat{N}$ does not depend on z within this interval. Consequently, the formal solution is given by the exponential of the sum of operators

$$\phi((n+1)z_c, x) = \exp\left(z_c(\hat{D} + \hat{N})\right)\phi(nz_c, x).$$

However, since $[\hat{D}, \hat{N}] \neq 0$, computing this exponential directly is non-trivial. We rely on the Trotter-Kato formula

$$\exp\left(z_c(\hat{D} + \hat{N})\right) = \lim_{M\to\infty}\left(\exp\left(\frac{z_c}{M}\hat{D}\right)\exp\left(\frac{z_c}{M}\hat{N}\right)\right)^M.$$

This formula justifies the numerical approximation: for a small step $h = z_c/M$, the total propagation is approximated by sequentially applying the operators $\exp(h\hat{D})$ and $\exp(h\hat{N})$. This is equivalent to solving two simplified differential equations sequentially on each sub-interval of length h.

Step 1: Diffraction (Application of $\exp(h\hat{D})$) First, we solve the homogeneous diffraction part $\partial_z\phi = \hat{D}\phi$. This is efficiently performed in the Fourier domain, where the operator $\hat{D}$ is diagonal. Let $\hat{\phi}(z, \kappa)$ be the Fourier transform with respect to the transverse variable x. The spatial derivative operator ∂_x^2 becomes a scalar multiplication by $-\kappa^2$:

$$\partial_z\hat{\phi}(z, \kappa) = -\frac{i\kappa^2}{2k}\hat{\phi}(z, \kappa).$$

The solution over a step h is obtained analytically in the frequency domain:

$$\hat{\phi}_{\text{diff}}(z + h, \kappa) = \hat{\phi}(z, \kappa)\exp\left(-\frac{ih\kappa^2}{2k}\right).$$

We then apply the inverse Fourier transform to recover the diffracted field in the spatial domain $\phi_{\text{diff}}(z + h, x)$.

Step 2: Refraction (Application of $\exp(h\hat{N})$) Second, we apply the effect of the random medium $\partial_z\phi = \hat{N}\phi$. Since $\hat{N}$ acts as a multiplicative potential in the spatial domain, the solution is

$$\phi(z + h, x) = \phi_{\text{diff}}(z + h, x)\exp\left(\frac{ikh}{2}\mu(z, x)\right).$$

By iterating these two steps from $z = 0$ to $z = L$, effectively generating a new realization of $\mu_n(x)$ every z_c, we reconstruct the transmitted wave profile through the random medium.

The method is summarized in Figure 1.

6 Time reversal for time-dependent waves in a random medium

In the previous sections, we analyzed the propagation and refocusing of monochromatic waves. To establish a more realistic and physically meaningful model, we now consider a time-dependent initial condition corresponding to an initial signal which has a flat spectrum over a frequency bandwidth $[\omega_0 - B, \omega_0 + B]$.

Due to the linearity of the wave equation, the broadband propagation and time-reversal process can be synthesized by superimposing the solutions of multiple independent monochromatic problems.

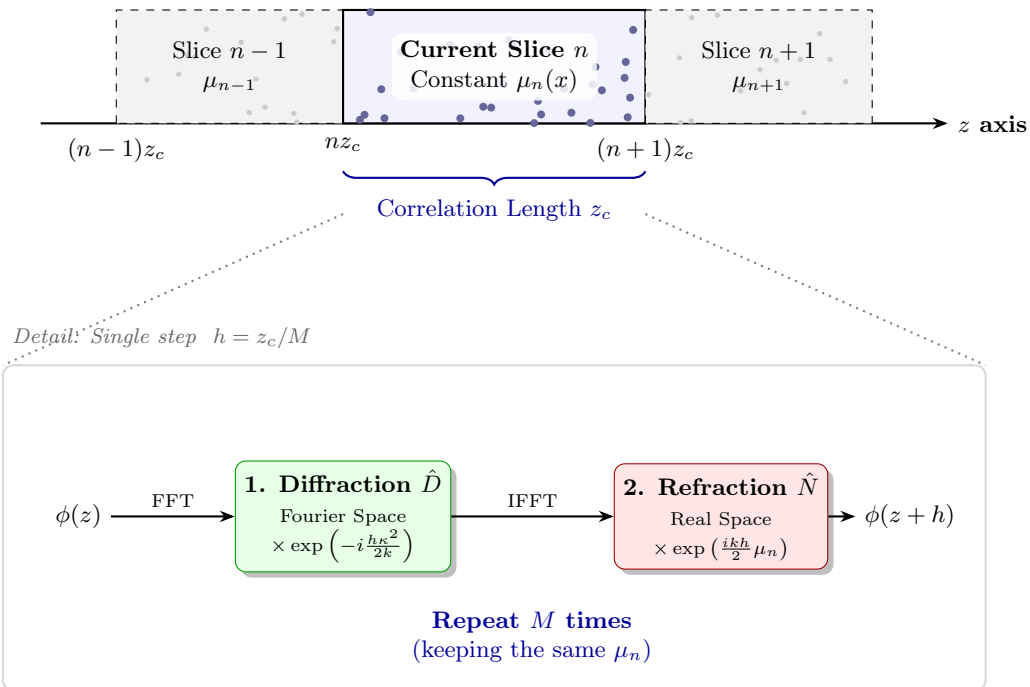

Figure 1: Schematic of the Split-Step Fourier Method in a random medium. The top layer shows the medium modeled as successive slices of length z_c, with random inhomogeneities. The bottom layer illustrates the iterative splitting algorithm applied within each slice.

6.1 Forward propagation and frequency discretization

Let us discretize the frequency bandwidth into N_{freq} evenly spaced components. The k-th discrete frequency is given by

$$\omega_k = \omega_0 + B\left(\frac{2k}{N_{\text{freq}}-1} - 1\right), \quad \text{for } k \in [\![0, N_{\text{freq}}-1]\!].$$

The frequency step is $\Delta\omega = \frac{2B}{N_{\text{freq}}-1}$.

For each fixed frequency ω_k, we compute the transmitted field at the mirror array ($z = L$) using the Split-Step Fourier method described in fig. 1. The complex spatial envelope is denoted by $\phi_{\omega_k}(L, x)$. The full wave function for ω_k is given

$$\hat{u}(L, x, \omega_k) = \phi_{\omega_k}(L, x) \exp\left(i\frac{\omega_k}{c_0}L\right).$$

6.2 Time reversal in the frequency domain

At the plane $z = L$, a time-reversal mirror records the signal and re-emits it. In the frequency domain, time reversal corresponds to complex conjugation. Taking into account the the mirror defined by $\chi_M(x)$, the initial condition for the back-propagating waee is

$$\widehat{u^{\text{tr}}}(L, x, \omega_k) = \overline{\hat{u}(L, x, \omega_k)}\chi_M(x) = \overline{\phi_{\omega_k}(L, x)}\chi_M(x) \exp\left(-i\frac{\omega_k}{c_0}L\right).$$

To simulate the back-propagation to the source plane (equivalent to propagating from $z = L$ to $z = 2L$ in an unfolded medium), we solve the paraxial wave equation again for each frequency. The time-reversal decomposes as

$$\widehat{u^{\text{tr}}}(z, x, \omega_k) = \phi^{\text{tr}}_{\omega_k}(z, x) \exp\left(i\frac{\omega_k}{c_0}(z - 2L)\right).$$

We denote the resulting envelope at the focal plane as $\phi^{\mathrm{tr}}_{\omega_k}(2L, x)$. The associated full spatial field is

$$\widehat{u^{\mathrm{tr}}}(2L, x, \omega_k) = \phi^{\mathrm{tr}}_{\omega_k}(2L, x).$$

6.3 Time-domain reconstruction

To reconstruct the full time-domain refocused signal $u^{\mathrm{tr}}(2L, x, t)$ around the focal point, we apply the inverse discrete Fourier transform. Summing over all frequency components yields

$$\begin{aligned} u^{\mathrm{tr}}(2L, x, t) &\simeq \frac{\Delta\omega}{2\pi} \sum_{k=0}^{N_{\mathrm{freq}}-1} \widehat{u^{\mathrm{tr}}}(2L, x, \omega_k) \exp(-i\omega_k t) \\ &= \frac{\Delta\omega}{2\pi} \sum_{k=0}^{N_{\mathrm{freq}}-1} \phi^{\mathrm{tr}}_{\omega_k}(2L, x) \exp\left(-i\omega_k t\right). \end{aligned}$$

A Usage of Generative AI

Generative AI was not employed for the derivation of mathematical formulas or the core theoretical development of this project. However, the Gemini was used for:

- Refining English phrasing and improving the overall narrative flow of the report.
- Troubleshooting code errors and discussing specific implementation details during the development phase.
- Generating the TikZ code used to create the illustration of fig. 1.